

Build-A-Thon X Unstop

Theme : Open Innovation (Meme Dating App)

LLMates : Ship Mode Activated ("Where Matching Internet Souls, One Meme at a Time")

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What the Project Does

It's an AI-powered meme dating application that matches users based on their interests, internet culture, humor preferences, and personality traits rather than relying solely on traditional dating metrics such as age, location, or profile photos.

The application collects user-selected interests through a Mood Board, analyzes the personality and online behavior patterns implied by those interests, and generates a compatibility report between two users.

The generated report includes:

- **Meme Personality Analysis**
- **Compatibility Score**
- **Shared Interests**
- **Match Explanation**
- **Meme-Based Pickup Line**
- **Personalized Icebreaker**

The entire matchmaking process is powered by a locally running Large Language Model (LLM), ensuring privacy and offline execution.

Problem Being Solved

Traditional dating platforms focus heavily on demographics, appearance, and location-based matching.

However, modern relationships are often built through:

- **Shared humor**
- **Meme culture**
- **Gaming communities**
- **Anime fandoms**
- **Technology interests**
- **Online behavior patterns**
- **Internet-native communication styles**

Two people may instantly connect because they understand the same meme, gaming reference, coding joke, or fandom culture.

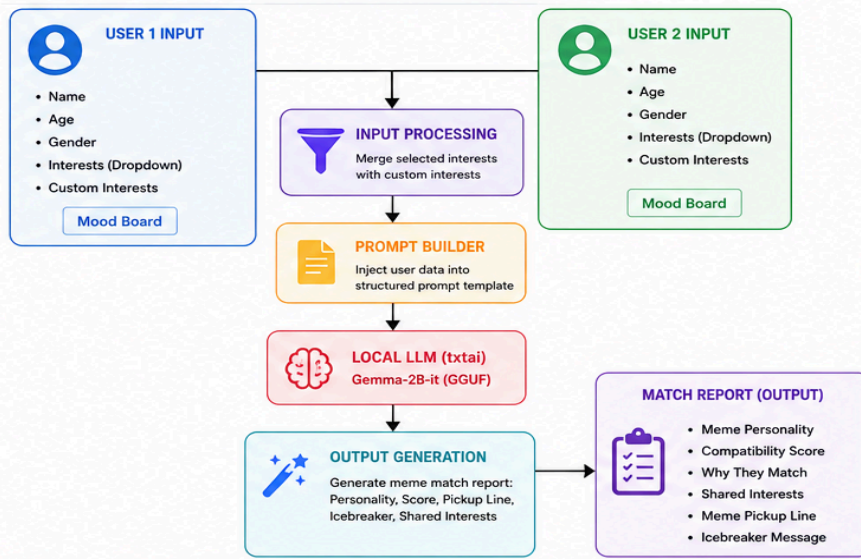
Existing matchmaking systems rarely capture these cultural and behavioral signals.

LLMates AI addresses this gap by:

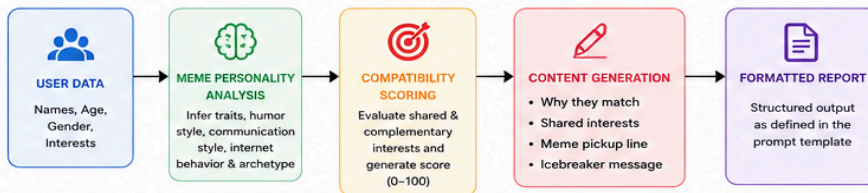
- Understanding user interests beyond demographics
- Inferring personality and communication style
- Detecting shared internet culture
- Generating fun and engaging matchmaking experiences

The result is a compatibility system that evaluates how people think, laugh, communicate, and interact online.

ARCHITECTURE / WORKFLOW DIAGRAM



PROMPT FLOW (INSIDE LLM)





4. TECHNOLOGY STACK

Category	Technology
Programming Language	Python
LLM	Gemma-2B-it (GGUF via txtai)
Frameworks	LangChain, txtai, Gradio
Interface	Gradio (Web UI)
Model Type	Local LLM (Offline Inference)
Prompting	Single Prompt Architecture

</> 6. SOURCE CODE EXPLANATION

Component / Function	Purpose
<code>TxtaiLangChainLLM</code> (Class)	Wraps txtai LLM as LangChain compatible model
<code>MOOD_BOARD_OPTIONS</code> (List)	Predefined list of common interests
<code>MEME_DATING_PROMPT</code> (String)	Main prompt for matchmaking
<code>run_matchmaking_ui()</code> (Function)	Core matchmaking logic
Gradio UI (Blocks)	User interface
<code>btn.click()</code>	Event trigger
<code>app.launch()</code>	Entry point



5. HOW IT WORKS (HIGH LEVEL)

1. Users enter their details and select mood board interests (or add custom ones).
2. The inputs are injected into a structured prompt.
3. The local LLM (Gemma 2B) analyzes both users.
4. It generates a meme match report with:
 - Meme personalities
 - Compatibility score
 - Why they match
 - Shared interests
 - Meme pickup line
 - Icebreaker message,



3. KEY FEATURES

- Meme Personality Analysis
- Compatibility Scoring (0–100)
- Shared Interests Detection
- Meme Pickup Line Generation
- Icebreaker Message Generation
- Custom Interests Support
- Local LLM (Offline, Private)
- Interactive Gradio Web UI

Future Improvements

Several enhancements can make LLMates AI more intelligent and scalable.

1. Embedding-Based Matching

Instead of relying solely on prompt reasoning:

- Generate vector embeddings
- Calculate similarity scores
- Improve compatibility accuracy

2. Multi-User Recommendations

Recommend:

- Top 5 matches
- Best meme partners
- Friendship matches

instead of comparing only two users.

3. User Profile Database

Store:

- Match history
- Interests
- Previous compatibility reports

for personalized recommendations.

4. Real-Time Chat Assistant

Add an AI wingman that:

- Suggests replies
- Generates memes
- Recommends conversation starters

5. Advanced Personality Modeling

Integrate:

- Big Five Personality Traits
- Behavioral Signals
- Emotional Compatibility

for deeper matchmaking.

6. Mobile & Cloud Deployment

Deploy as:

- Web Application
- Android App
- iOS App

for larger user adoption.

Note :- Will support Multiple Languages

Limitations

Although effective for demonstrations and prototypes, the system has several limitations.

1. LLM-Dependent Results

Compatibility depends heavily on the reasoning quality of the language model.

Different models may produce different outputs.

2. No Historical Learning

The system does not learn from previous matches.

Each matchmaking session is independent.

3. Limited Interest Taxonomy

The predefined Mood Board contains a limited set of interests.

Users may require additional categories.

Conclusion

LLMates AI demonstrates how Large Language Models can be used to create a new generation of dating experiences centered around internet culture, shared humor, fandoms, and personality-driven interactions.

Instead of asking *"Do these people look compatible?"*, the system explores a more modern question:

"Would these two people laugh at the same memes?"